

|                                                     |                                   |                                    |
|-----------------------------------------------------|-----------------------------------|------------------------------------|
| <b>STUDENT<br/>FULL NAME:</b>                       | <b>STUDENT<br/>NUMBER:</b>        |                                    |
| <b>SUBJECT:</b><br>Professional Development/Methods | <b>TEST DATE:</b><br>20 JULY 2020 | <b>DURATION:</b><br>TAKE HOME TEST |



TEST 2

## FACULTY OF INFORMATICS & DESIGN

COURSE:

Advanced Diploma in Information Technology

SUBJECT COORDINATOR:

Mr T. Makhurane

LECTURERS / EXAMINERS:

Mr T. Makhurane

MODERATOR:

Internal:

External:

| Question | Student Mark | Max. Score |
|----------|--------------|------------|
| 1        |              | 20         |
| 2        |              | 20         |
| 3        |              | 20         |
| 4        |              | 20         |
| 5        |              | 20         |
|          |              |            |
|          |              |            |
|          |              |            |
|          |              |            |
|          |              |            |
|          |              |            |
|          |              | 100        |

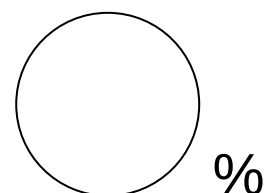
### SPECIAL INSTRUCTIONS:

☛ Answer all the questions that are written in bold.  
Hand-in question paper at the end of examination

### REQUIREMENTS:

☛ Students should not leave within the first hour even if they have completed.

☛ Hand-in question paper together with the answer book at the end of examination



**QUESTION 1. [20]**

|          |                 |
|----------|-----------------|
| Interest | 10%             |
| Year     | Cash Flow (ZAR) |
| 0        | -120000         |
| 1        | 34000           |
| 2        | 34000           |
| 3        | 34000           |
| 4        | 34000           |
| 5        | 34000           |

1.1 Calculate the Net Present Value. [7]

1.2 Calculate the payback period. [1]

1.3 [9]

|      |                 |
|------|-----------------|
| Year | Cash Flow (ZAR) |
| 0    | -500000         |
| 1    | 200000          |
| 2    | 300000          |
| 3    | 200000          |

**Table 1**

Calculate the Internal Rate of Return for the situation in Table 1.

**QUESTION 2. [20]**

2.1 Explain the different levels of Rationale Management. [5]

2.2 Explain the purpose of Rationale Management and explain how it contributes to successful project management. [8]

2.3 What information, or data, must be captured to facilitate Rationale Management? [7]

**QUESTION 3. [20]**

3.1 What is software configuration management? [4]

3.2 Describe the workings of a suitable tool for software configuration management. [8]

3.3 Describe how the use of an organized approach to software configuration management can contribute toward the successful management of a software development project. [8]

**QUESTION 4. [20]**

4.1 What the major differences between a waterfall software development lifecycle and Unified Process model? [10]

4.2 It is your task to develop a lifecycle model for a large software system intended to support the functions of a municipal council. This software system will include functionality related to deeds management, driver and vehicle licensing, zoning of districts, electricity distribution and payments among other functions. [10]

Explain, with justification, the details of a lifecycle model for a task of this magnitude.

**QUESTION 5. [20]**

| Activity | Immediate Predecessors | Optimistic | Most Likely | Pessimistic |
|----------|------------------------|------------|-------------|-------------|
| A        | -                      | 42         | 50          | 55          |
| B        | A                      | 15         | 32          | 42          |
| C        | A                      | 41         | 43          | 47          |
| D        | B                      | 64         | 78          | 88          |
| E        | B                      | 40         | 45          | 58          |
| F        | C, D                   | 2          | 4           | 7           |
| G        | E                      | 12         | 17          | 23          |
| H        | E, F                   | 20         | 24          | 25          |

- 5.1 Draw the network. [6]
- 5.2 Determine the expected finish time. [6]
- 5.3 Calculate the time by which we have a 95% confidence of finishing. [4]
- 5.4 Calculate the probability of finishing 2 days after the expected finish time. [4]

## Addenda

Standardised Normal Distribution (z-table) (Full Table)

| z   | 0.00   | 0.01   | 0.02   | 0.03   | 0.04   | 0.05   | 0.06   | 0.07   | 0.08   | 0.09   |
|-----|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| 0.0 | 0.5000 | 0.5040 | 0.5080 | 0.5120 | 0.5160 | 0.5199 | 0.5239 | 0.5279 | 0.5319 | 0.5359 |
| 0.1 | 0.5398 | 0.5438 | 0.5478 | 0.5517 | 0.5557 | 0.5596 | 0.5636 | 0.5675 | 0.5714 | 0.5753 |
| 0.2 | 0.5793 | 0.5832 | 0.5871 | 0.5910 | 0.5948 | 0.5987 | 0.6026 | 0.6064 | 0.6103 | 0.6141 |
| 0.3 | 0.6179 | 0.6217 | 0.6255 | 0.6293 | 0.6331 | 0.6368 | 0.6406 | 0.6443 | 0.6480 | 0.6517 |
| 0.4 | 0.6554 | 0.6591 | 0.6628 | 0.6664 | 0.6700 | 0.6736 | 0.6772 | 0.6808 | 0.6844 | 0.6879 |
| 0.5 | 0.6915 | 0.6950 | 0.6985 | 0.7019 | 0.7054 | 0.7088 | 0.7123 | 0.7157 | 0.7190 | 0.7224 |
| 0.6 | 0.7257 | 0.7291 | 0.7324 | 0.7357 | 0.7389 | 0.7422 | 0.7454 | 0.7486 | 0.7517 | 0.7549 |
| 0.7 | 0.7580 | 0.7611 | 0.7642 | 0.7673 | 0.7704 | 0.7734 | 0.7764 | 0.7794 | 0.7823 | 0.7852 |
| 0.8 | 0.7881 | 0.7910 | 0.7939 | 0.7967 | 0.7995 | 0.8023 | 0.8051 | 0.8078 | 0.8106 | 0.8133 |
| 0.9 | 0.8159 | 0.8186 | 0.8212 | 0.8238 | 0.8264 | 0.8289 | 0.8315 | 0.8340 | 0.8365 | 0.8389 |
| 1.0 | 0.8413 | 0.8438 | 0.8461 | 0.8485 | 0.8508 | 0.8531 | 0.8554 | 0.8577 | 0.8599 | 0.8621 |
| 1.1 | 0.8643 | 0.8665 | 0.8686 | 0.8708 | 0.8729 | 0.8749 | 0.8770 | 0.8790 | 0.8810 | 0.8830 |
| 1.2 | 0.8849 | 0.8869 | 0.8888 | 0.8907 | 0.8925 | 0.8944 | 0.8962 | 0.8980 | 0.8997 | 0.9015 |
| 1.3 | 0.9032 | 0.9049 | 0.9066 | 0.9082 | 0.9099 | 0.9115 | 0.9131 | 0.9147 | 0.9162 | 0.9177 |
| 1.4 | 0.9192 | 0.9207 | 0.9222 | 0.9236 | 0.9251 | 0.9265 | 0.9279 | 0.9292 | 0.9306 | 0.9319 |
| 1.5 | 0.9332 | 0.9345 | 0.9357 | 0.9370 | 0.9382 | 0.9394 | 0.9406 | 0.9418 | 0.9429 | 0.9441 |
| 1.6 | 0.9452 | 0.9463 | 0.9474 | 0.9484 | 0.9495 | 0.9505 | 0.9515 | 0.9525 | 0.9535 | 0.9545 |
| 1.7 | 0.9554 | 0.9564 | 0.9573 | 0.9582 | 0.9591 | 0.9599 | 0.9608 | 0.9616 | 0.9625 | 0.9633 |
| 1.8 | 0.9641 | 0.9649 | 0.9656 | 0.9664 | 0.9671 | 0.9678 | 0.9686 | 0.9693 | 0.9699 | 0.9706 |
| 1.9 | 0.9713 | 0.9719 | 0.9726 | 0.9732 | 0.9738 | 0.9744 | 0.9750 | 0.9756 | 0.9761 | 0.9767 |
| 2.0 | 0.9772 | 0.9778 | 0.9783 | 0.9788 | 0.9793 | 0.9798 | 0.9803 | 0.9808 | 0.9812 | 0.9817 |
| 2.1 | 0.9821 | 0.9826 | 0.9830 | 0.9834 | 0.9838 | 0.9842 | 0.9846 | 0.9850 | 0.9854 | 0.9857 |
| 2.2 | 0.9861 | 0.9864 | 0.9868 | 0.9871 | 0.9875 | 0.9878 | 0.9881 | 0.9884 | 0.9887 | 0.9890 |
| 2.3 | 0.9893 | 0.9896 | 0.9898 | 0.9901 | 0.9904 | 0.9906 | 0.9909 | 0.9911 | 0.9913 | 0.9916 |
| 2.4 | 0.9918 | 0.9920 | 0.9922 | 0.9925 | 0.9927 | 0.9929 | 0.9931 | 0.9932 | 0.9934 | 0.9936 |
| 2.5 | 0.9938 | 0.9940 | 0.9941 | 0.9943 | 0.9945 | 0.9946 | 0.9948 | 0.9949 | 0.9951 | 0.9952 |
| 2.6 | 0.9953 | 0.9955 | 0.9956 | 0.9957 | 0.9959 | 0.9960 | 0.9961 | 0.9962 | 0.9963 | 0.9964 |
| 2.7 | 0.9965 | 0.9966 | 0.9967 | 0.9968 | 0.9969 | 0.9970 | 0.9971 | 0.9972 | 0.9973 | 0.9974 |
| 2.8 | 0.9974 | 0.9975 | 0.9976 | 0.9977 | 0.9977 | 0.9978 | 0.9979 | 0.9979 | 0.9980 | 0.9981 |
| 2.9 | 0.9981 | 0.9982 | 0.9982 | 0.9983 | 0.9984 | 0.9984 | 0.9985 | 0.9985 | 0.9986 | 0.9986 |
| 3.0 | 0.9987 | 0.9987 | 0.9987 | 0.9988 | 0.9988 | 0.9989 | 0.9989 | 0.9989 | 0.9990 | 0.9990 |
| 3.1 | 0.9990 | 0.9991 | 0.9991 | 0.9991 | 0.9992 | 0.9992 | 0.9992 | 0.9992 | 0.9993 | 0.9993 |
| 3.2 | 0.9993 | 0.9993 | 0.9994 | 0.9994 | 0.9994 | 0.9994 | 0.9994 | 0.9995 | 0.9995 | 0.9995 |
| 3.3 | 0.9995 | 0.9995 | 0.9995 | 0.9996 | 0.9996 | 0.9996 | 0.9996 | 0.9996 | 0.9996 | 0.9997 |
| 3.4 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9998 |
| 3.5 | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 |
| 3.6 | 0.9998 | 0.9998 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 |

**Formulas**

|                   |                                                                                        |
|-------------------|----------------------------------------------------------------------------------------|
| Free Float        | $FF(i, j) = E(j) - E(i) - D_{ij}$                                                      |
| Independent Float | $IF(i, j) = \begin{cases} 0 \\ E(j) - L(i) - D_{ij} \end{cases} \quad IF(i, j) \geq 0$ |
| Total Float       | $TF(i, j) = L(j) - E(i) - D_{ij}$                                                      |
| Expected Duration | $ED = \frac{A + 4M + B}{6}$                                                            |
| Variance          | $\left( \frac{B - A}{6} \right)^2$                                                     |
|                   | $z = \frac{x - \mu}{s}$                                                                |
|                   | $x = zs + \mu$                                                                         |